Daniel Lo

AI Robotics Engineer | Robotics / AI Application Engineer

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Robotics control, computer vision, ROS 2, backend systems, and cross-domain integration

Summary

AI Robotics Engineer at AMTRAN TECHNOLOGY CO., LTD. with an M.S. in Electrical Engineering. I connect Isaac Sim, ROS 2, MoveIt 2, computer vision, and robot control into verifiable real-world workflows, and also ship integrated products with FastAPI, Flutter, PostgreSQL, and Docker.

Experience

AMTRAN TECHNOLOGY CO., LTD. | AI Robotics Engineer

May 2025 - Present | New Taipei, Taiwan

- Built Isaac Sim, ROS 2, MoveIt 2, and cuMotion environments for pose control, motion planning, collision handling, and object manipulation.
- Extended the control architecture from UR5 to TM5S through configuration-driven robot and gripper integration.
- Integrated RealSense, DOPE, AprilTag, and OpenCV for 3D pose estimation, TF conversion, hand-eye calibration, and grasp-target calculation.
- Developed YOLO OBB HDMI insertion and AprilTag HDMI / power-cable unplugging workflows integrating TM Flow, TM API, ROS 2, and gripper control.
- Developed a PatchCore anomaly-detection API / UI with ROI extraction, preprocessing, template replacement, and false-positive analysis.
- Built VR control, imitation-learning, and digital-twin data pipelines integrating images, robot state, gripper state, actions, and Rotation 6D.

ISCOM | Software Engineer Intern

Feb 2022 - Jun 2022 | Kaohsiung, Taiwan

- Participated in backend API development, frontend integration, and web feature implementation.
- Worked with C#, ASP.NET MVC, JavaScript, jQuery, MS SQL, and Git in a collaborative environment.

Selected Projects

YOLO OBB HDMI Insertion System

Robot control / visual servoing / integration | TM5S, TM Flow, TM API, ROS 2, YOLO OBB, OpenCV

Used YOLO OBB to detect HDMI-hole position and orientation, then controlled a TM5S wrist camera for alignment and insertion.

Evidence: Real-robot workflow, camera geometry, and multi-axis control

Card Guide: Credit Card Reward Comparison PWA

Product / API / deployment | Flutter, FastAPI, PostgreSQL, PWA, Docker

A live Taiwan credit-card reward comparison product with official-source disclosure and rule validation.

Evidence: 2026-08-08: 22 Flutter tests, backend pytest 40 passed, Web release, and HTTP smoke passed

[Open live demo]

Multi-stack Docker Self-hosted Service Platform

DevOps / platform operations / security boundary | Docker, Nginx, Tailscale, route registry

Managed private multi-stack services with deployment scripts, route allowlists, health checks, and rollback paths.

Evidence: Route / Nginx syntax, Compose config, health checks, and HTTP smoke verified

Core Skills

Robotics	Isaac Sim, Isaac ROS, ROS 2, MoveIt 2, cuMotion, TM5S / UR5	Computer Vision	OpenCV, YOLO OBB, DOPE, AprilTag, RealSense, PatchCore
Backend	Python, FastAPI, SQLAlchemy, Alembic, PostgreSQL, Docker	Frontend	Flutter, Dart, PWA, HTML / CSS, PyQt / PyQt6
DevOps	Docker Compose, Nginx, Tailscale, GitHub Actions, Linux	Data / AI	HDF5, imitation learning, digital twins, Rotation 6D, NVIDIA Isaac GR00T

Education & Honors

- M.S. in Electrical Engineering, National Taiwan Normal University | Sep 2022 - Jul 2024
- B.S. in Computer and Communication, National Pingtung University | Sep 2018 - Jun 2022
- RHCSA; TQC+ C Programming; FIRA Roboworld Cup First Place; National Technical Creativity Competition Third Place

Portfolio & Links

Portfolio: https://daniel-locy.github.io/

Engineering projects: https://daniel-locy.github.io/engineering/

GitHub: https://github.com/Daniel-LoCY

LinkedIn: https://www.linkedin.com/in/chung-yung-lo-790ab6330/

Verification snapshot

- The portfolio includes a live Flutter / FastAPI product with dated test evidence and a public demo.
- The Docker platform documents route allowlists, reproducibility checks, health checks, smoke tests, and rollback paths.
- Private repositories and internal infrastructure details are intentionally excluded from this public resume.